

TOMSK, Russian Federation

WMO: 294300

Lat: 56.500N

Lon: 84.917E

Elev: 457

StdP: 14.46

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.6	-28.0	-38.6	0.6	-33.2	-33.1	0.9	-27.7	14.9	10.5	13.1	8.8	1.1	230	0.562

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.5	83.7	67.3	80.8	65.7	77.9	64.3	69.8	79.2	68.0	77.3	66.3	74.7	3.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.6	99.6	73.9	64.7	93.1	72.4	63.0	87.5	70.8	34.2	79.3	32.7	77.4	31.3	74.9	77.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.4	8.9	7.4	DB	-37.8	89.1	6.4	2.5	-42.4	90.9	-46.1	92.3	-49.7	93.7	-54.3	95.5	(4)
(5)			WB	-37.4	72.5	6.5	2.1	-42.1	74.0	-45.9	75.2	-49.6	76.3	-54.3	77.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	34.3	-1.1	6.2	20.9	36.6	50.7	62.1	65.9	60.7	49.0	35.8	17.5	5.2		
	DBStd	25.46	15.60	13.27	11.23	9.73	9.91	7.95	5.82	6.66	8.05	9.18	14.21	15.15		
	HDD50	7183	1585	1226	903	413	116	9	0	5	115	447	974	1390		
	HDD65	11404	2050	1646	1368	853	452	143	60	166	482	905	1424	1855		
	CDD50	1443	0	0	0	10	138	373	494	335	86	7	0	0		
	CDD65	189	0	0	0	0	10	57	89	31	2	0	0	0		
	CDH74	1619	0	0	0	2	158	513	669	250	27	0	0	0		
	CDH80	333	0	0	0	0	39	107	137	47	3	0	0	0		
(14) Wind	WSAvg	3.1	3.1	3.4	3.5	3.8	3.6	2.7	2.4	2.3	2.4	2.9	3.4	3.3		
(15) Precipitation	PrecAvg	20.7	1.2	0.7	0.8	1.2	1.7	2.4	2.6	3.0	1.8	2.0	1.9	1.5		
	PrecMax	29.1	2.6	1.8	1.8	3.1	3.6	5.2	5.9	6.2	5.6	4.1	3.4	3.0		
	PrecMin	14.9	0.4	0.0	0.1	0.3	0.4	0.7	0.0	0.6	0.5	0.6	0.4	0.5		
	PrecStd	3.0	0.5	0.4	0.4	0.7	0.7	1.1	1.3	1.5	1.1	0.8	0.7	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.3	34.4	48.0	69.2	84.9	87.0	87.3	85.4	78.0	63.9	45.2	33.8		
		MCWB	30.5	31.2	39.3	52.9	63.9	66.1	70.1	69.2	63.2	51.2	41.6	31.7		
	2%	DB	27.6	31.0	42.7	62.4	78.2	83.0	83.6	80.2	71.0	58.0	40.6	31.3		
		MCWB	26.3	28.8	35.6	49.5	59.3	65.6	68.7	66.5	58.9	49.0	38.0	30.2		
	5%	DB	24.5	27.8	39.1	56.1	73.3	79.9	80.6	76.4	66.1	53.0	37.2	28.3		
		MCWB	23.1	26.0	33.4	45.8	56.5	64.6	67.1	64.4	56.5	47.1	35.2	27.5		
	10%	DB	20.8	24.5	35.9	51.1	67.8	76.6	77.8	72.7	61.4	48.1	34.1	25.0		
		MCWB	19.6	23.0	31.9	43.1	54.3	63.2	65.6	62.6	54.1	43.7	32.6	24.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	30.6	32.3	40.2	54.2	65.6	70.3	72.8	70.7	64.1	53.5	42.9	32.4		
		MCDB	31.7	33.7	47.0	66.8	82.7	81.1	82.3	81.5	74.3	59.0	44.5	33.2		
	2%	WB	26.6	29.0	36.4	50.4	61.3	67.9	70.7	68.2	60.1	50.3	38.4	30.2		
		MCDB	27.7	30.8	41.1	60.8	74.4	78.3	79.9	77.3	69.1	55.9	39.9	31.0		
	5%	WB	23.2	26.0	34.1	46.6	58.1	66.0	69.0	66.1	57.4	47.2	35.3	27.4		
		MCDB	24.4	27.6	38.0	54.9	70.0	76.5	77.6	73.8	64.2	52.9	37.0	28.4		
	10%	WB	19.7	23.0	32.2	43.3	55.3	64.3	67.4	63.9	55.0	44.1	32.7	24.0		
		MCDB	20.8	24.5	35.6	50.6	66.1	74.0	75.6	70.9	60.5	48.0	33.9	25.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.8	15.4	17.6	17.1	19.7	18.3	17.5	17.0	15.8	12.1	10.8	12.1		
		MCDBR	12.5	13.4	19.2	24.6	27.3	24.0	21.8	22.9	23.8	20.3	10.7	12.5		
		MCWBR	12.4	11.8	13.5	14.3	13.3	10.2	9.4	10.6	12.4	12.4	8.7	12.2		
	5% WB	MCDBR	12.8	12.7	16.2	23.3	24.8	21.1	19.3	20.2	21.4	18.9	10.5	12.5		
		MCWBR	12.8	11.6	12.0	14.4	13.6	10.1	9.5	10.5	11.8	12.1	8.9	12.3		
(40) Clear-Sky Solar Irradiance	taub		0.292	0.325	0.374	0.401	0.407	0.406	0.436	0.408	0.371	0.351	0.319	0.293		
	taud		1.994	1.966	1.911	2.062	2.218	2.304	2.216	2.285	2.335	2.238	2.050	1.950		
	Ebn at Noon		174	218	238	255	264	266	253	253	246	213	161	134		
	Edh at Noon		21	32	45	45	41	38	41	36	30	25	20	17		
(44) All-Sky Solar Radiation	RadAvg		248	563	1005	1393	1582	1833	1669	1341	881	436	205	146		
	RadStd		26	37	70	108	158	186	112	106	124	36	34	27		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page